$m\ddot{x} + kx = 0$ $\quad (*)_1$ $\quad$ 2nd order 1D

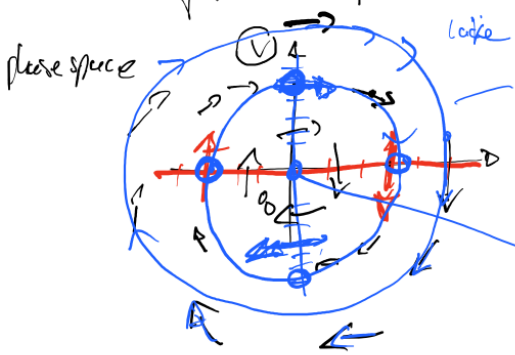
{Force balance from Newton} $\quad$ 1st order 2D system

phase space : state of m : $\left.\begin{aligned} \dot{x} &= v \\ \dot{v} &= -\frac{k}{m}x \end{aligned}\right\} (*)_1$

$\omega^2 = \frac{k}{m}$ $\quad \Rightarrow \quad \begin{aligned} \dot{x} &= v \\ \dot{v} &= -\omega^2 x \end{aligned} \quad \hat{=} \quad \dot{x} = Ax \quad \vec{x} = \begin{pmatrix} x \\ v \end{pmatrix}$ position velocity

$\Leftrightarrow \begin{pmatrix} \dot{x} \\ \dot{v} \end{pmatrix} = \begin{pmatrix} v \\ -\omega^2 x \end{pmatrix} \quad \hat{=}$ vector field

vector of velocities for state $(x, v)^T$

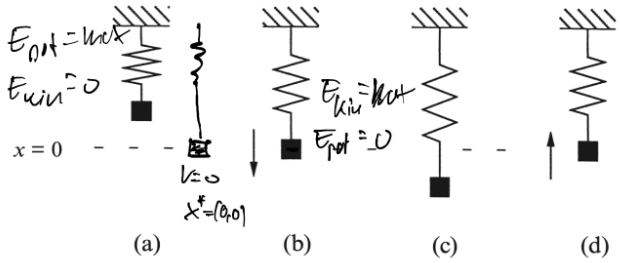


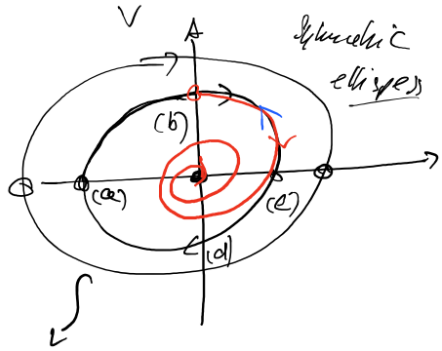
$v = 0 \quad \begin{pmatrix} \dot{x} \\ \dot{v} \end{pmatrix} = \begin{pmatrix} 0 \\ -\omega^2 x \end{pmatrix}$

$x = 0 \quad \begin{pmatrix} \dot{x} \\ \dot{v} \end{pmatrix} = \begin{pmatrix} v \\ 0 \end{pmatrix}$

$\boxed{x^* = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \text{ fixed point}}$

closed orbits = periodic oscillations





Hyperbolic ellipses

ellipses correspond to a specific energy level

$$\boxed{C = \text{Energy} = \underbrace{\omega^2 x^2}_{\text{potential}} + \underbrace{v^2}_{\text{kinetic Energy}}}$$

Example 2:

$$\dot{x} = Ax \qquad A = \begin{pmatrix} a & 0 \\ 0 & -1 \end{pmatrix}$$

Graph the phase space portrait as a varies.

from $-\infty \ldots +\infty$

$$\boxed{\dot{x} = \underbrace{\begin{pmatrix} a & 0 \\ 0 & -1 \end{pmatrix}}_{} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax \\ -y \end{pmatrix}}$$

$\rightarrow \dot{x} = ax$
$\rightarrow \dot{y} = -y$ } matrix multiplication

$x \xrightarrow{a}$ $\quad y \xrightarrow{-1}$

no interaction $\Rightarrow$ states are independent of each other.

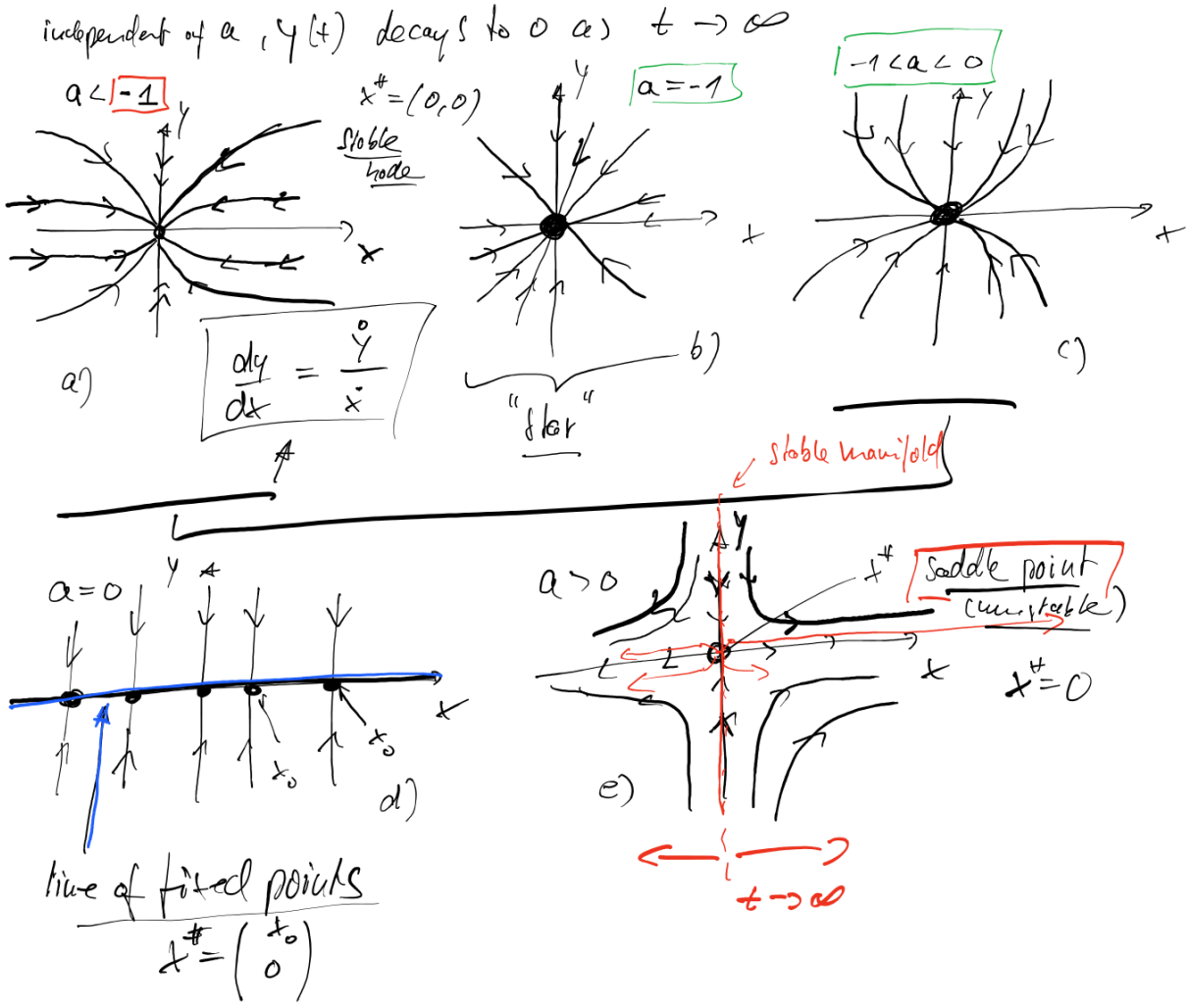
= uncoupled states

Solution:

$$x(t) = x_0 e^{\boxed{a}t} \overset{a=0}{=} x_0 \qquad \dot{x} = a x_0 e^{at} = ax$$

$$y(t) = y_0 e^{\boxed{-1}t} \qquad \dot{y} = -y_0 e^{-t} = -y$$

independent of a, $y(t)$ decays to 0 as $t \to \infty$



Stability language

→ $x^* = 0$ is an attracting fixed point

all trajectories that start close by will approach x^* $t \to \infty$

$$\boxed{x(t) \longrightarrow x^* \text{ as } t \to \infty}$$

if this holds for all trajectories

{Liapunov stable}

x^* is <u>unstable</u> if neither attracting nor Liapunov stable

<u>Classification of linear systems</u>

$\dot{x} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} x = Ax$ stable, unstable, <u>star? orbit? spiral?</u>

need to study this matrix

$\rightarrow$ Δ determinant and trace τ of A

1 step $\rightarrow$ general solution λt ← matrix

$(*)_3$ $\vec{x}(t) = e^{\lambda t} \vec{v}$. $\vec{v} \neq 0$ some fixed vector we need to figure.

"exponential growth/decay along $\vec{v}$"

<u>2nd step</u>: condition for $\vec{v}$, λ fic

$\vec{x}(t)$ into $(*)_3$

$$\lambda e^{\lambda t} \vec{v} = e^{\lambda t} A \vec{v}$$

$$\{ \dot{x} = Ax \}$$

$\lambda \vec{v} = A \vec{v}$ "Eigenproblem" $\circledast_4$

multiply — matrix — eigenvector

eigenvalues

$(\lambda \mathbb{1} - A)\vec{v} = 0 \quad | -1$

$\{ \lambda \vec{v} - A \vec{v} = 0 \}$?

3rd step: characteristic equation $\circledast_4$

$$\det(A - \lambda \mathbb{1}) = 0$$

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$\det \begin{pmatrix} a-\lambda & b \\ c & d-\lambda \end{pmatrix} = 0$$

$$\longrightarrow \boxed{\lambda^2 - \tau \lambda + \Delta = 0}$$

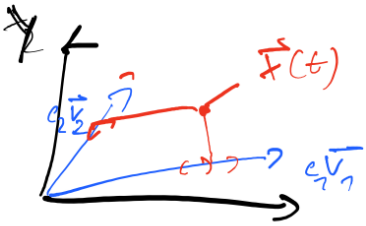
$\tau = \text{trace}(A) = a + d$

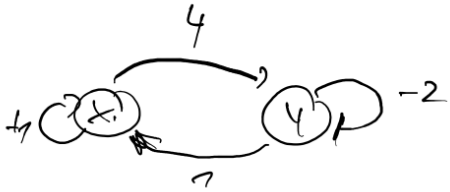
$\Delta = \det(A) = ad - bc$

$$r_1 = \frac{\tau + \sqrt{\tau^2 - 4\Delta}}{2} \qquad r_2 = \frac{\tau - \sqrt{\tau^2 - 4\Delta}}{2}$$

$r_1 \neq r_2$ typical solution $\Rightarrow$ $\vec{v}_1$, $\vec{v}_2$ eigenvectors that correspond to r_1, r_2

$$\boxed{\vec{x}(t) = c_1 e^{r_1 t} \vec{v}_1 + c_2 e^{r_2 t} \vec{v}_2}$$





Example 3:

$$A = \begin{pmatrix} 1 & 1 \\ 4 & -2 \end{pmatrix}$$

$\dot{x} = x + y$

$\dot{y} = 4x - 2y$

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 4 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

① $\tau = -1 \qquad \Delta = -6$

$r^2 + r - 6 = 0 \Rightarrow \boxed{\begin{matrix} r_1 = 2 \\ r_2 = -3 \end{matrix}}$ eigenvalues ②

③ eigenvectors: $\left(\begin{matrix} 1-r & 1 \\ 4 & -2-r \end{matrix}\right) \left(\begin{matrix} v_1 \\ v_2 \end{matrix}\right) = \left(\begin{matrix} 0 \\ 0 \end{matrix}\right)$ $\qquad \vec{v} = \left(\begin{matrix} v_1 \\ v_2 \end{matrix}\right)$

{ solve eigenproblem by plugging in $r_1, r_2 \Rightarrow$ determine v_1, v_2

$r_1 = 2$: $\left(\begin{matrix} -1 & 1 \\ 4 & -4 \end{matrix}\right) \left(\begin{matrix} v_1 \\ v_2 \end{matrix}\right) = \left(\begin{matrix} 0 \\ 0 \end{matrix}\right) \rightarrow \left(\begin{matrix} v_1 \\ v_2 \end{matrix}\right) = \boxed{\left(\begin{matrix} 1 \\ 1 \end{matrix}\right) = \vec{v}_1}$

$r_2 = -3$: $\left(\begin{matrix} 4 & 1 \\ 4 & 1 \end{matrix}\right) \left(\begin{matrix} v_1 \\ v_2 \end{matrix}\right) = \left(\begin{matrix} 0 \\ 0 \end{matrix}\right) \rightarrow \left(\begin{matrix} v_1 \\ v_2 \end{matrix}\right) = \boxed{\left(\begin{matrix} 1 \\ -4 \end{matrix}\right) = \vec{v}_2}$

(4) general solution: $\vec{x}(t) = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} 1 \\ -4 \end{pmatrix} e^{-3t}$

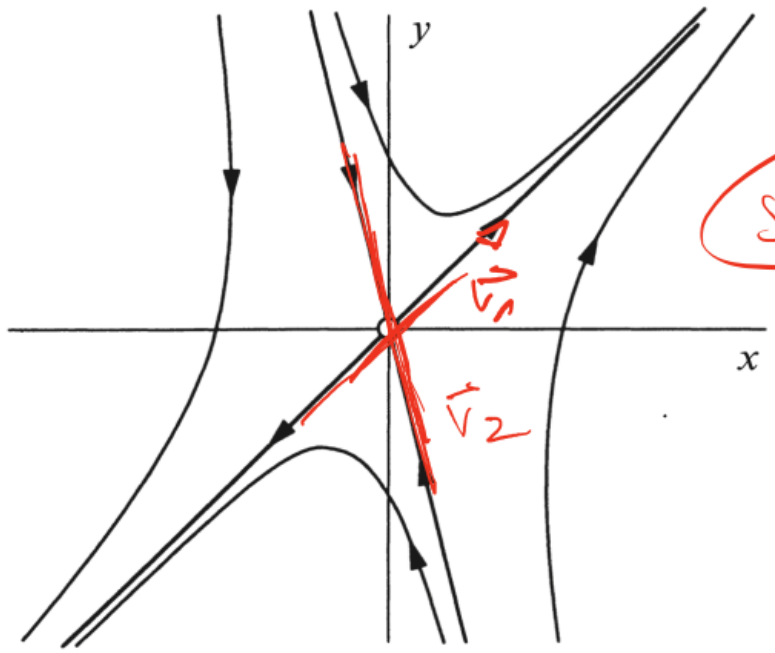
(5) c_1, c_2: $(x_0, y_0) = (2, -3)$ at $t = 0$

$\rightarrow \begin{pmatrix} 2 \\ -3 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ -4 \end{pmatrix}$

$\rightarrow c_1 = 1, \; c_2 = 1$

(6) $$\boxed{\begin{aligned} x(t) &= e^{2t} + e^{-3t} \\ y(t) &= e^{2t} - 4e^{-3t} \end{aligned}}$$

$\rightarrow$ What does $\vec{x}(t)$ look like



saddle point